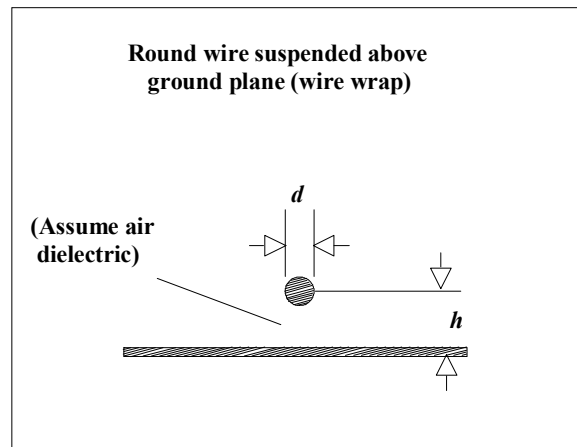


Formulas included in this spreadsheet:

Round wire characteristic impedance	ZROUND()
Round wire propagation delay	PROUND()
Round wire inductance	LROUND()
Round wire capacitance	CROUND()

Variables used:

d Diameter of wire (in.)
 h Height of wire above ground (in.)
 x Length of wire (in.)
 (We assume the wire is suspended in air, for which the relative dielectric constant is 1.00.)



round

Characteristic impedance of round wire above ground plane (Ω):

$$ZROUND(d,h) := 60 \cdot \ln\left(\frac{4 \cdot h}{d}\right)$$

Propagation delay per in. of round wire above ground plane (s/in):

$$PROUND(d,h) := 84.72 \cdot 10^{-12} \quad (\text{assume air dielectric})$$

Inductance of round wire above ground plane (H):

$$LROUND(d,h,x) := x \cdot 5.08 \cdot 10^{-9} \cdot \ln\left(\frac{4 \cdot h}{d}\right)$$

Capacitance of round wire above ground plane (F):

$$CROUND(d,h,x) := \left(\frac{x \cdot 1.413 \cdot 10^{-12}}{\ln\left(\frac{4 \cdot h}{d}\right)} \right)$$

Example round wire calculations

Diameter of AWG 30 wire (in.)	D:= .01
Length of wire (in.)	X:= 2.000
Height above ground (in.)	H:= .100

Characteristic impedance (Ω):

$$\text{ZROUND}(D,H) = 221.333$$

Total inductance (H):

$$\text{LROUND}(D,H,X) = 3.748 \times 10^{-8}$$

Same result in nH:

$$\text{LROUND}(D,H,X) \cdot 10^9 = 37.479$$

Inductance per in. (H):

$$\text{LROUND}(D,H,1) = 1.874 \times 10^{-8}$$

Total capacitance (F):

$$\text{CROUND}(D,H,X) = 7.661 \times 10^{-13}$$

Same result in units pF:

$$\text{CROUND}(D,H,X) \cdot 10^{12} = 0.766$$

Capacitance per in. (F):

$$\text{CROUND}(D,H,1) = 3.83 \times 10^{-13}$$